Nuclear Deterrence in the Euro-Atlantic Area

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Abstract

Deterrence succeeds or fails in the mind of the adversary. This paper examines strategic nuclear deterrence as a behavioural-science concept, with reference to Russia's war against Ukraine and its renewed importance for the United Kingdom. It first reviews the classical framework of capability, credibility and communication, before considering research showing that perception, cognitive bias and reference-dependent attitudes to risk mediate every deterrent threat. It then proposes two analytical lenses. *Entropic deterrence* describes the deterrent effect of disorder within a strategic system, which complicates an adversary's calculations but can also be exploited, and can draw an unwise actor into inadvertent escalation. *Shear-force failure* describes the sudden, severe breakdown of a deterrent after incremental pressure, such as Russian hybrid activity, that offers few warning signs. These lenses are combined in a four-phase diagnostic framework and illustrated through two UK decisions of 2025. The paper concludes that uncertainty is double-edged: posture should remain calculable and reassuring for allies while being difficult for an adversary to exploit.

Keywords: nuclear deterrence; strategic stability; prospect theory; misperception; escalation; hybrid warfare; Russia; NATO; United Kingdom

1. Introduction

Deterrence seeks to prevent an action by persuading an adversary that its likely costs and risks will outweigh its expected benefits (Freedman, 2004; Mazarr, 2018). In its simplest form, this appears to be a calculation: a state considers whether aggression will succeed, what response it may provoke, and whether the result is worth the danger. In practice, however, the calculation is rarely simple or objective. Leaders act under pressure, with incomplete information and through institutional processes that can distort, delay or reinterpret information. The decisive variable is therefore not merely the military capability a state possesses, but the adversary's *perception* of that capability and of the willingness to use it. Deterrence is consequently a behavioural problem as much as a military one (Jervis et al., 1985; Mazarr, 2018).

Nuclear weapons make this problem particularly acute. Their destructive power can give even a small possibility of use political significance, yet their very severity may make threats to use them less credible in some circumstances. Nuclear deterrence has therefore long occupied a central place in British defence thinking: it concerns not only the possession of a strategic capability, but also how that capability is understood by allies, adversaries and domestic audiences.

Russia's full-scale invasion of Ukraine in February 2022, accompanied by repeated nuclear signalling, has given these questions renewed urgency (Williams et al., 2023). Russia has sought to deter direct Western intervention while pursuing conventional war against Ukraine. NATO members, meanwhile, have needed to support Ukraine and reinforce the Alliance's eastern flank without creating an unnecessary path to direct conflict with Russia. This is a setting in which

*The views expressed in this paper are the author's own. They do not represent the policy or internal thinking of His Majesty's Government or the Ministry of Defence, and the paper draws only on published sources.

messages may be deliberately ambiguous, accidentally unclear, or interpreted differently by different audiences.

This paper asks two questions. First, how can policymakers distinguish disorder that complicates an adversary's planning from disorder that an adversary can exploit, or that may draw either side into inadvertent escalation? Second, how should they think about the possibility that a deterrent may break suddenly, with few warning signs, after a period of incremental pressure, rather than weakening visibly over time? To answer them, the paper first outlines the established theory of deterrence and the behavioural evidence that qualifies it (§ 2), and sets out the contemporary Euro-Atlantic context (§ 3). It then introduces two analytical lenses, entropic deterrence and shear-force failure, and combines them in a four-phase framework intended to organise judgement rather than predict outcomes (§ 4). The framework is used to interpret two UK decisions of 2025 as illustrative cases (§ 5). Finally, the paper considers the framework's limits (§ 6) and its implications for UK and allied policy (§ 7).

2. Theoretical foundations

2.1. The requirements of deterrence

The earliest nuclear-era theorists emphasised that a deterrent threat is only as strong as the adversary's belief in it. Kaufmann (1954) argued that deterrence requires not only the capability to inflict unacceptable costs, but also the adversary's conviction that the deterring state would be willing to do so. A threat that cannot be carried out is hollow; equally, a threat that can be carried out but is not believed is unlikely to shape behaviour.

This insight is commonly expressed through three interdependent requirements. The first is *capability*: the practical means to deny an attack or impose unacceptable costs in response. The second is *credibility*: the adversary's belief that those means would actually be employed under the circumstances in question. The third is *communication*: the successful conveyance of commitments, interests and possible consequences in terms the adversary can understand. It is useful to separate a fourth requirement, *comprehension*, from communication. A message has little deterrent value if it is received but interpreted in a different way from that intended.

These requirements work together rather than independently. Capability without credibility may be dismissed as a bluff. Credibility without communication may be misread or overlooked. Communication without capability may expose weakness rather than deter. Deterrence is therefore better understood as multiplicative than additive: a serious failure in one element can weaken the effect of the others. This point matters for the rest of the paper, because uncertainty may affect each element differently. It may conceal useful capabilities from an adversary, but it may also obscure political resolve or confuse an ally about the commitment on which it relies.

2.2. Denial, punishment and compellence

The classical literature distinguishes two principal ways in which deterrence operates (Snyder, 1959, 1961; Mazarr, 2018). *Deterrence by denial* seeks to persuade an adversary that aggression will not achieve its objectives. It rests on defensive capability, resilience, dispersal, readiness and the restriction of opportunity. If an attacker expects its operation to fail, or to produce only limited gains at unacceptable cost, it may decide not to begin.

Deterrence by punishment, by contrast, seeks to persuade an adversary that aggression may initially succeed but will bring severe costs afterwards. Those costs can be military, economic or political; at the nuclear level, they may include the possibility of nuclear retaliation. Punishment is therefore often associated with retaliatory threats, while denial is associated with making the original act impracticable or unprofitable.

Denial is often regarded as more reliable because it places a tangible obstacle in an aggressor's path. Forces deployed forward, for example, can both impede an attack and demonstrate that other states are already committed to a response. Threats of later punishment may be more difficult to assess because an aggressor may doubt that the defender will accept the costs and risks of retaliation (Mazarr, 2018). Nevertheless, denial and punishment are not alternatives in every case. A credible posture can combine both: denial may reduce an adversary's prospects of success, while punishment can make even a limited success unattractive.

Schelling (1966) distinguishes both forms of deterrence from *compellence*. Deterrence seeks to stop an adversary from taking an action; compellence seeks to induce it to take an action, or reverse one already taken. This distinction is important because compellence is generally harder. A state being compelled must visibly change its behaviour under pressure, and may resist because compliance risks domestic humiliation, loss of status or further demands.

For Schelling, coercion is not simply the possession of military force. It is a bargaining process in which timing, clarity, proportionality and the perceived willingness to accept risk all matter. His idea of the "threat that leaves something to chance" (Schelling, 1960) captures the fact that a degree of uncontrollable danger can itself be coercive. A state may not specify exactly what it will do, but may make clear that continued escalation creates risks that neither side can fully control. Figure 1 places deterrence within this wider coercive toolkit.

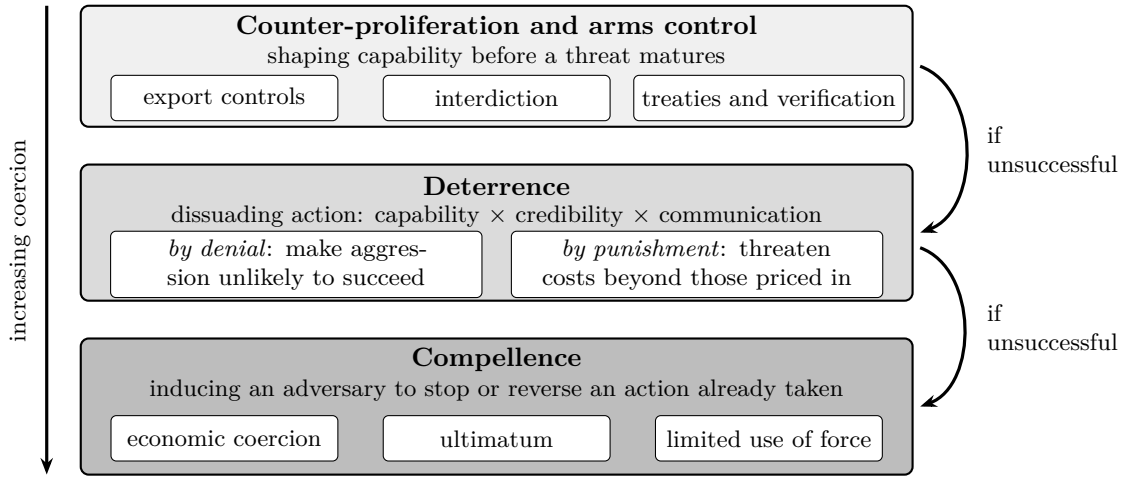


Figure 1. A schematic of the coercive toolkit, ordered by increasing coercion. Counter-proliferation and arms control shape capability before a threat matures; deterrence seeks to dissuade action; compellence seeks to stop or reverse an action already taken. The arrows indicate movement to the next tier when the previous one fails.

The figure should not be read as a mechanical escalatory ladder. States can use several instruments at once, and may move backwards as well as forwards. Its purpose is to show that deterrence occupies the space before an action is taken, while compellence becomes more relevant when deterrence has not prevented an action and policymakers seek to alter the adversary's behaviour.

2.3. Perception and misperception

The central behavioural difficulty is that adversaries do not interpret signals objectively. Decision-makers receive information through existing beliefs, assumptions and political incentives. Jervis (1976) showed that leaders tend to assimilate new information to prior expectations and to project their own assumptions onto opponents. A deployment intended as reassurance may therefore be read as preparation for attack; an ambiguous message intended to create caution may be taken as evidence of weakness.

Historical cases show that aggressors have repeatedly acted against apparently stronger opponents because they discounted the defender's resolve or believed that they had no acceptable alternative (Jervis et al., 1985; Lebow and Stein, 1989). Such errors are not always the product of irrationality. An adversary may possess information unavailable to outside observers, face intense domestic constraints, or calculate that the political consequences of inaction are worse than the military risks of action. The problem for deterrence is that a defender's own assessment of a threat's clarity is not enough. What matters is the meaning that a particular adversary attaches to it.

Lebow's analysis of Argentina's decision to invade the Falkland Islands in 1982 is particularly relevant to the United Kingdom (Lebow, 1985). Argentina did not simply ignore British signals; rather, its leaders interpreted them through assumptions about British interests, domestic politics and likely resolve. The case illustrates why possession of military capability does not automatically produce deterrence. A deterrent posture is only as effective as the specific adversary perceives its capability, credibility and communication to be (Mazarr, 2018).

2.4. Prospect theory and the risk of the losing actor

Prospect theory helps explain why deterrence can fail even when a threat appears compelling under a conventional cost–benefit calculation. Kahneman and Tversky (1979) showed that individuals tend to evaluate choices relative to a reference point rather than according to final outcomes alone. They also tend to be risk-averse when protecting gains but more willing to accept risk when attempting to avoid or recover losses.

Applied to international relations, this suggests that leaders who believe they are losing territory, status, security or regime survival may take risks that they would reject in more favourable circumstances. Levy (1997) argues that decision-makers may become risk-acceptant when they frame their position as one of loss. The relevant question is therefore not only whether an adversary faces high objective costs, but whether it believes restraint would confirm an unacceptable loss.

This matters especially in the nuclear domain. It does not mean that a state facing conventional defeat will necessarily use nuclear weapons. It does mean that the actor most likely to contemplate extreme escalation may be the one that believes its conventional position, status or survival is already deteriorating. Deterrence policy must therefore pay close attention to adversary perceptions of loss, not merely to material balances of power.

2.5. The measurement problem

A final difficulty is that deterrence is inherently hard to evaluate. Successful deterrence produces a non-event. The absence of an attack cannot, by itself, establish whether the adversary was deterred, never intended to attack, lacked the capability to do so, or was restrained by some other consideration (Lebow and Stein, 1990). Conversely, a challenge to a deterrent does not necessarily prove that deterrence has failed completely; it may reveal an adversary testing a boundary it believes remains below the threshold of response.

Assessment should therefore avoid simple success-or-failure judgements based on one event. It should instead triangulate evidence: changes in military capability, political commitments, force posture, alliance cohesion, adversary rhetoric, intelligence indicators and observed behaviour. This does not remove uncertainty, but it makes analysis more disciplined. It also provides the foundation for the diagnostic framework introduced in § 4.

3. The contemporary Euro-Atlantic context

These theoretical issues have renewed practical significance in the Euro-Atlantic area. Strategic stability between NATO and Russia is under greater strain than at any time since the end of the Cold War. The Intermediate-Range Nuclear Forces Treaty ended in August 2019. Russia suspended its participation in New START in February 2023, and the treaty expired on 5 February 2026 without a successor, leaving the strategic arsenals of the United States and Russia without binding limits for the first time in more than fifty years (United Nations Secretary-General, 2026). The resulting environment contains fewer formal constraints and fewer established channels through which the largest nuclear powers can provide reassurance or manage mistrust.

Russia's stated nuclear doctrine has also evolved. The revised *Basic Principles* of November 2024 state that aggression by a non-nuclear state with the participation or support of a nuclear state will be regarded as a joint attack (President of the Russian Federation, 2024). Whether or not such declaratory policy predicts operational behaviour, it affects the environment in which allied decision-makers interpret Russian red lines and Russia interprets Western support for Ukraine.

Russian strategic thought treats deterrence as an integrated, cross-domain activity extending across nuclear, conventional, cyber, information and other non-military instruments in peace and war (Bruusgaard, 2016). Since 2022, Russian nuclear rhetoric has had at least two connected purposes: discouraging direct Western military intervention and seeking to divide NATO support for Ukraine (Williams et al., 2023). This is not a return to a simple Cold War model of two stable and clearly separated blocs. Instead, competition takes place across several domains, with frequent hybrid activity below the level of declared armed conflict (Mazarr, 2015).

Technological developments add further complexity. Inexpensive one-way attack drones have exposed the vulnerability of concentrated, high-value assets and have increased the practical importance of dispersal, resilience and air defence. At the same time, rapid communication and open-source reporting can amplify the political effects of military movements and make signalling more difficult to control. For the United Kingdom, experience in the South Atlantic in 1982 and in post-Cold War operations in the Middle East offers useful lessons, but only limited guidance for a nuclear-armed confrontation involving Russia and NATO.

The central challenge is therefore not simply to possess capable forces. It is to sustain a posture that reassures allies, discourages adversaries, limits opportunities for miscalculation and remains credible during a crisis. The next section develops two lenses intended to help structure that assessment.

4. Two analytical lenses and a phase framework

The contemporary environment increases the value of a structured way to assess deterrent effects. The following concepts are not established terms in the literature. They are offered as analytical extensions of the behavioural findings reviewed above, and as aids to practical judgement rather than predictive models.

The first concept concerns disorder. Every deterrent operates within a system of actors, institutions and information flows that is never fully ordered, and the resulting unpredictability can itself deter. Yet the same disorder gives an adversary room to manoeuvre, and it can lead an unwise actor into steps whose consequences neither side can calculate. The second concept concerns the manner in which deterrence fails. Under sustained, incremental pressure, a posture may appear stable until it breaks suddenly and severely, with few reliable warning signs.

4.1. Entropic deterrence

Entropic deterrence denotes the deterrent effect of disorder within a strategic system. The term is used by analogy with entropy, a measure of disorder in a physical system and of uncertainty in an informational one: not as a precise quantity, but to describe the unpredictability an adversary faces when attempting to plan aggression. The system in question includes the defender's own alliance, with its many governments, institutions and decision processes; the adversary's command structures; and an information environment in which events are reported, amplified and misread at speed. The greater the disorder in this system, the harder it is for an adversary to judge that a limited action can be undertaken safely, because detection, attribution, allied reactions, military consequences and escalation risks all become harder to forecast.

Some of this disorder is deliberate. Calculated ambiguity about thresholds and possible responses can complicate planning and discourage opportunistic action, consistent with Schelling's idea of a threat that leaves something to chance (Schelling, 1960). A defender may reasonably avoid setting out an exact, automatic response to every conceivable provocation. Excessive precision can create exploitable gaps just below a stated threshold, and explicit threats risk a "commitment trap" in which the defender must either carry out a threat it would prefer not to execute or accept a loss of credibility (Sagan, 2000).

Much of the disorder, however, is not the defender's to control. It arises from friction, disagreement, bureaucratic delay, differing national interests and imperfect coordination within an alliance. Such disorder may still deter if an adversary fears that events could produce a stronger response than expected. But it also creates opportunity. An adversary can hide hostile activity in the noise, exploit gaps between allies' assessments and decision timelines, and use deniable methods that make attribution slow and contested. Allied disorder therefore acts on the adversary's calculations in two directions at once: it complicates aggression, but it may also persuade the adversary that resolve is divided and that political seams can be exploited. This tension echoes the alliance security dilemma described by Snyder (1984), in which allies' competing fears of abandonment and entrapment shape their commitments to one another; entropic deterrence considers how that internal friction reaches the adversary's calculations.

The greatest danger lies with an actor that behaves unwisely inside a system it cannot calculate. Where neither side can be confident where the other's thresholds lie, a step intended as a limited probe may cross one unknowingly; equally, a defender may react to an ambiguous event with a response the adversary had not priced in. Disorder can therefore produce inadvertent escalation, or a sudden, high-cost response, without either side having chosen it (Posen, 1991). Studies of complex, tightly coupled systems reach a similar conclusion: interactions that no participant anticipated can produce accidents that no participant intended (Perrow, 1984; Sagan, 1993).

This distinguishes entropic deterrence from strategic ambiguity. Ambiguity is a declaratory choice made by one actor; entropy is a property of the whole system, much of which neither side controls. The analytical task is therefore to distinguish the disorder a defender chooses to create from the disorder it merely suffers. Useful disorder is directed outward, towards the adversary's ability to plan aggression. Harmful disorder spreads inward, weakening allied confidence, obscuring the basic political commitments on which deterrence depends, or offering the adversary seams to exploit. A successful posture does not remove all ambiguity; it manages ambiguity so that it complicates aggression without making allied commitments unintelligible.

4.2. Shear-force failure

Shear-force failure describes a deterrent that breaks suddenly and severely, with few reliable warning signs. The term is borrowed by analogy from structural engineering. In a reinforced-concrete beam, for example, failure in bending is ductile: large deflections and visible cracking

give warning well before collapse. Failure in shear is brittle: once capacity is exceeded, it can occur abruptly, with little visible distress beforehand, which is why designers ensure that a member would fail in bending first. The analogy is imperfect, but useful. A deterrent posture may absorb repeated pressure, probing and rhetoric without visible distress, and then fail abruptly and disproportionately when one further increment exceeds a threshold that neither side had clearly identified.

Russian hybrid activity illustrates the mechanism. Cyber operations, sabotage, airspace incursions, coercive rhetoric, disinformation and other grey-zone methods can each be calibrated to fall below the threshold that would trigger a decisive response (Mazarr, 2015; Bruusgaard, 2016). This is the incrementalism that Schelling (1966) called salami tactics, and it presents the defender with a "boiling frog" problem: because each step is small, none appears to justify a strong reaction, and the cumulative change in the strategic situation is hard to see from inside it. Each step may also test political cohesion and willingness to accept risk, not only military readiness.

When failure comes, it can take two forms, and both are hard to forecast. Credibility may collapse: a visible failure to act, a public disagreement between allies or an unexpectedly weak response may lead the adversary to conclude that a red line was never real, and to act more boldly. Alternatively, the defender may respond suddenly and forcefully, because cumulative provocation has exhausted its political tolerance or because delay has raised the cost of inaction. Either way, the break is disproportionate to the event that triggers it, and restoring stability afterwards may require more effort, risk and expenditure than maintaining it beforehand.

The concept also engages a debate in the behavioural literature. Press (2005) finds that leaders judge the credibility of threats mainly by the current balance of power and interests, rather than by the defender's past behaviour. On that view, a single unanswered probe need not destroy credibility. Shear-force failure is consistent with this finding where incremental pressure has gradually changed the adversary's assessment of the defender's current interests and cohesion: the collapse reflects an accumulated shift in the adversary's calculation, revealed by a single event, rather than a reputation lost in one episode.

Shear-force failure should not be interpreted as inevitable, or as an argument for responding forcefully to every challenge. It is instead a warning that the signs of approaching failure will be weak, and that apparently minor events can acquire broader meaning when they are interpreted against a history of prior signals and assumptions. Policymakers should therefore track the cumulative pattern of an adversary's activity, not only the scale of each action, and consider what each event is likely to mean for perceptions of resolve.

4.3. Phases of deterrence

The two lenses can be combined in a four-phase diagnostic framework, illustrated schematically in figure 2. The framework is intended to support judgements about the "health" of a deterrent posture and the adjustment it may require. It is not a model that predicts behaviour, and the phases are not mutually exclusive or permanently sequential. A state may occupy different phases in relation to different adversaries, domains or geographical areas at the same time.

The value of the framework lies in prompting policymakers to ask what kind of problem they are facing. Is the priority to establish a new deterrent? To maintain an existing posture? To reinforce it after a test? Or to restore stability after deterrence has already failed? In each phase, assessment should triangulate observable changes in capability, political commitments, alliance coordination and adversary behaviour. No single indicator can establish success or failure.

Establish. A credible deterrent is built where none previously existed, or where an existing arrangement is judged insufficient for a new threat. Foundational capabilities are acquired, legal

and alliance frameworks are put in place, and red lines or commitments are communicated so as to shape the adversary's baseline calculations. The challenge is to make the new posture sufficiently visible to be understood, while avoiding commitments so rigid that they can be easily tested or circumvented.

Maintain. The deterrent holds and strategic stability is preserved. Activity and investment are calibrated to sustain capability, reassure partners and demonstrate persistent resolve without provoking an unnecessary arms race or escalatory spiral. Maintenance is not passive. It includes exercising forces, renewing political commitments, improving resilience and ensuring that allies continue to understand both the purpose and limits of the posture.

Reinforce. The adversary tests thresholds, for example through grey-zone activity, coercive signalling or conventional mobilisation (point A in figure 2). This is the phase in which incremental pressure accumulates, and in which the warning signs of shear-force failure are weakest. The aim is to prevent the adversary from interpreting restraint as weakness or discovering an exploitable gap. Rapid but proportionate action may be required to strengthen capability, restate political commitment and signal, publicly or privately, that red lines are close to being crossed. Reinforcement should address the specific inference the adversary may be drawing, rather than simply increasing activity for its own sake.

Restore. The deterrent has broken (point B), typically through shear-force failure, and the situation is escalating. When dissuasion alone can no longer prevent or halt the action, the response may shift towards compellence in Schelling's sense: applying pressure to induce the adversary to reverse course or take another specified action. This phase concerns both the immediate response to deterrence failure and the longer effort to create a stable deterrent again (point C). It does not imply that restoring the previous status quo is always feasible, desirable or risk-free.

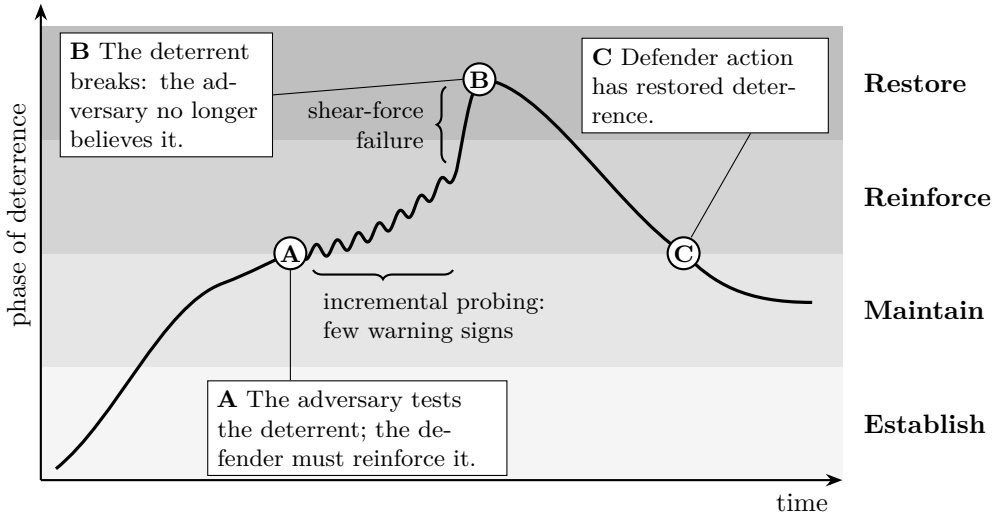


Figure 2. A schematic of the phases of deterrence over time. The adversary tests the deterrent (A) and then applies incremental pressure, each step small enough to give few clear warning signs. The near-vertical rise to B marks shear-force failure: the deterrent breaks suddenly rather than eroding visibly. The longer descent from B to C indicates that restoring deterrence takes more time and effort than maintaining it. The vertical axis is ordinal: it indicates the phase, and hence the intensity of effort required, rather than a measured quantity.

The diagram emphasises that the move from reinforcement to restoration may be abrupt, and that the period before it may look deceptively stable. It also shows why early recognition of incremental testing matters. If policymakers can reinforce a posture before an adversary

concludes that a commitment is empty, they may avoid the higher costs associated with restoring deterrence after open failure.

5. Application: the United Kingdom's posture in 2025

The framework can be used to interpret two UK decisions of 2025. The purpose here is illustrative rather than causal. The available public evidence does not establish that either decision was a direct response to a particular adversary probe, nor that either decision produced a measurable deterrent outcome. They nevertheless demonstrate how capabilities, alliance arrangements and communication may interact within a broader deterrent posture.

In June 2025, the government announced the purchase of twelve F-35A aircraft to join NATO's dual-capable aircraft nuclear mission, describing the decision as the biggest strengthening of the UK's nuclear posture in a generation (Prime Minister's Office, 2025a). The decision reintroduced a nuclear role for the Royal Air Force for the first time since the withdrawal of the WE 177 free-fall bomb in 1998 (Imperial War Museums, n.d.; Prime Minister's Office, 2025a). It also added an air-delivered option, within NATO's nuclear-sharing arrangements, alongside the United Kingdom's independent submarine-based Trident deterrent.

The significance of the decision is not simply numerical. An air-delivered capability can broaden the range of options available to NATO and may make allied planning more credible in circumstances where an adversary doubts that strategic nuclear forces would be employed. It also connects the United Kingdom more visibly to NATO's shared nuclear mission. From the perspective of the framework, this can be read as a reinforcing measure: it adds capability, signals alliance participation and potentially affects an adversary's calculation of the risks associated with coercion against NATO. In shear-force terms, such measures add margin: by making commitments harder to doubt, they reduce the chance that a single probe reveals a deterrent to be hollow.

In July 2025, the Northwood Declaration with France stated that the two countries' nuclear forces "are independent, but can be coordinated", that "there is no extreme threat to Europe that would not prompt a response by our two nations", and that a UK–France Nuclear Steering Group would be established to coordinate nuclear policy, capabilities and operations (Prime Minister's Office, 2025b). The declaration was significant because the United Kingdom and France are Europe's two nuclear-armed states. It sought to make clear that national independence did not preclude coordination in the face of an extreme threat.

Viewed through the lens of entropic deterrence, these measures could have two effects at once. For allies, more visible capability and coordination may reduce harmful disorder by making commitments and planning arrangements more legible. For Moscow, the same measures may create useful uncertainty about which capability, and whose, might be relevant to a particular provocation. This is an interpretation of their possible signalling effect, not a demonstrated outcome or necessarily their sole declared purpose.

The distinction is important. A posture that is deliberately difficult for an adversary to calculate should not become obscure to the allies expected to rely on it. Additional capabilities and new coordinating bodies can strengthen deterrence only if their relationship to existing commitments is understood. Otherwise, complexity could create the kind of unintended disorder that entropic deterrence identifies as potentially harmful, and that an adversary could seek to exploit. The policy challenge is to communicate sufficient coherence to allies while retaining enough flexibility that an adversary cannot confidently identify a safe route to coercion.

6. Discussion

The behavioural perspective exposes both the value and the limits of deterrence. It cautions against treating adversaries as unitary, fully rational actors that mechanically compare costs and benefits. Decisions are instead taken by leaders and institutions under stress, with incomplete information and under domestic political pressure. A leadership that frames its position as one of loss may accept risks that a conventional rational-actor model would exclude (Lebow and Stein, 1989; Levy, 1997).

This has an important policy implication. Deterrence should not be assessed only in terms of whether a defender has communicated a threat clearly. Policymakers must also ask how an adversary is likely to interpret that threat given its strategic culture, domestic politics, military position and sense of loss or gain. The answer may differ across adversaries and may change during a crisis. A message that deterred yesterday may be insufficient when the same adversary believes that it has fewer acceptable alternatives.

The framework also highlights an ethical and operational dilemma. Deliberate ambiguity may deter by making aggression difficult to plan, but it can also lead adversaries or allies to misunderstand intent. Excessive clarity can be equally problematic if it creates rigid thresholds that invite action just below them. Effective policy therefore requires a balance: uncertainty where it constrains aggression, and clarity where it reduces the risk of accidental escalation or allied miscalculation.

Ukraine illustrates both the failure and the continuing relevance of deterrence. Deterrence did not prevent Russian aggression against a state outside NATO. At the same time, deterrence has so far helped to prevent direct military conflict between Russia and the Alliance across the boundary protected by Article 5 of the North Atlantic Treaty. This does not prove that deterrence has been effortless or complete. Rather, it demonstrates the importance of *intra-war deterrence*: efforts to limit the geographical scope, means and escalation of an ongoing conflict even after deterrence has failed in another respect.

The proposed framework has clear limitations. Its phases are ideal types, not sharply bounded states. Policymakers may disagree about whether an event is a test requiring reinforcement or a failure requiring restoration. Shear-force failure, in particular, is easier to identify after the event than to predict: if its warning signs were clear, the failure would not be sudden. Like all deterrence assessments, the framework remains subject to the measurement problem described in § 2. It cannot establish what an adversary believes with certainty, nor can it substitute for intelligence, military analysis or political judgement. Its value lies instead in structuring questions that might otherwise be overlooked: what disorder is useful, what disorder is damaging or exploitable, what signal is an adversary likely to infer, and whether a deterrent is being tested before its credibility changes more abruptly.

7. Conclusion

Nuclear deterrence has returned to the forefront of Euro-Atlantic security. Its effects are psychological as well as material, depending on capability, credibility, communication and the adversary's interpretation of all three. Classical deterrence theory remains essential, but behavioural research demonstrates that credibility is specific to the adversary concerned and to that adversary's reference frame. A state operating from a perceived loss may become risk-acceptant precisely when restraint is most needed.

The concepts of entropic deterrence and shear-force failure extend this insight. Entropic deterrence draws attention to disorder within the strategic system: it can deter by making aggression incalculable, but it also gives an adversary room to exploit, and it can draw an unwise actor

into inadvertent escalation or a sudden, high-cost response. Shear-force failure highlights that a deterrent may break suddenly and severely after incremental pressure that offered few warning signs, which is what hybrid campaigns are designed to exploit. The four-phase framework offers a way to distinguish the tasks of establishing, maintaining, reinforcing and restoring deterrence.

Two implications follow for HMG and its allies. First, the behavioural assumptions underpinning deterrence policy should be tested through realistic crisis exercises that reflect time pressure, incomplete information, alliance disagreement, incremental hybrid pressure and the possibility of loss-framed decision-making. Such exercises should test not only military plans but also how messages will be interpreted by allies and adversaries.

Second, calculability should be treated as a designed feature of deterrent posture. Policymakers should be explicit about what allies need to understand, what an adversary should be unable to calculate confidently, and what ambiguity is deliberate rather than accidental. A posture intended to be difficult for Moscow to predict must nevertheless remain intelligible, credible and recoverable for the allies that depend upon it.

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